

WEEK 4 REPORT - Time-Series Visualization & Financial Dashboard

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INTRODUCTION:

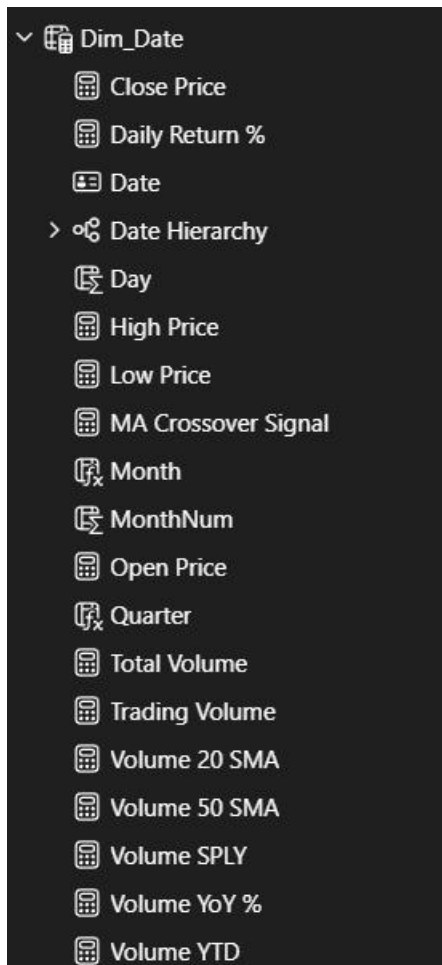
In Week 4, the Shopify stock data was explored in Power BI to understand how the stock price and trading volume changed over time. DAX measures were written to calculate moving averages, and interactive visuals were built to study the trends. Slicers and a custom tooltip were added so the dashboard is easy to explore and understand.

DATA PREPARATION:

The Shopify stock dataset, with Open, High, Low, Close and Volume columns, was loaded into Power BI. A separate date table (Dim_Date) was connected to the stock data through the Date column. This allowed the data to be analyzed across different time periods and made the moving-average calculations possible.

DAX MEASURES:

Three DAX measures were created for this analysis. The Volume 20 SMA and Volume 50 SMA measures calculate the average trading volume over the last 20 and 50 days. The Moving Average Crossover Signal compares the short-term and long-term averages and shows a "Bullish" or "Bearish" result. These measures help identify short-term and long-term trends in the data.



TRADING VOLUME ANALYSIS:

Trading volume was studied using the 20-Day and 50-Day Simple Moving Averages. The area chart shows how the average volume rises and falls from 2015 to 2025. High-activity periods appear around 2017, 2020 and 2022, and volume becomes lower and steadier after 2022. A second chart compares the daily Total Volume with the 20-Day SMA. Daily volume has many sudden spikes, while the moving average gives a smoother view of the overall trend. This makes it easier to tell short one-day spikes apart from real changes in market activity.

```
Trading Volume = SUM(Fact_Stock[volume])
```

```
Volume SPLY =  
CALCULATE(  
    [Trading Volume],  
    SAMEPERIODLASTYEAR(Dim_Date[Date])  
)
```

```

Volume 20 SMA =
VAR CurrentDate = MAX(Dim_Date[Date])
VAR PeriodDays = 20
VAR DateInPeriod =
    DATESINPERIOD(Dim_Date[Date], CurrentDate, -PeriodDays, DAY)
RETURN
CALCULATE(AVERAGE(Fact_Stock[Volume]), DateInPeriod)

```

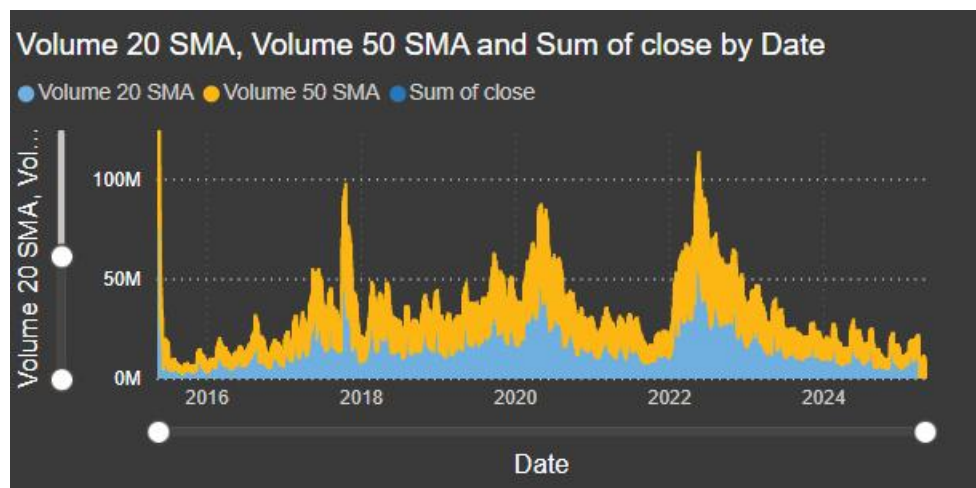
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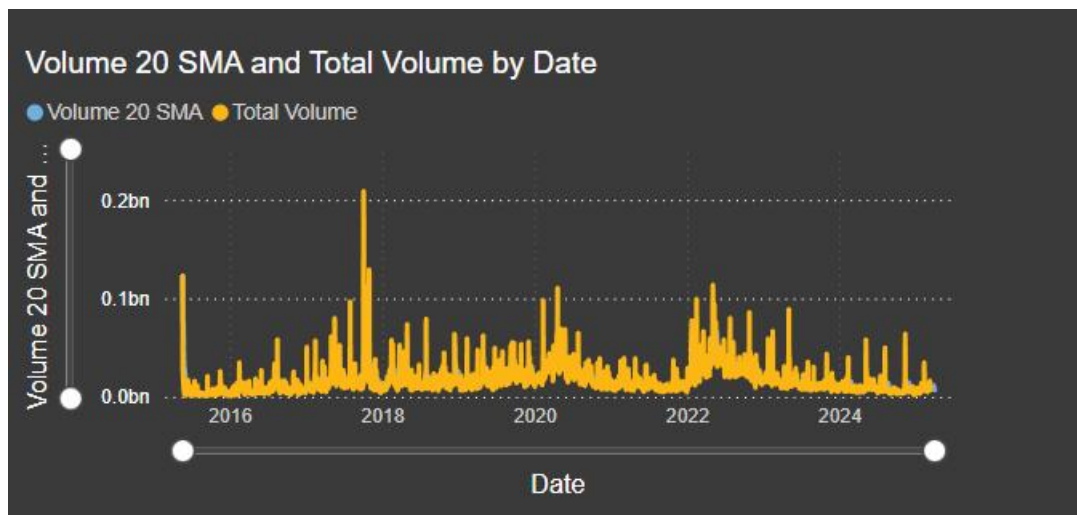
Volume 50 SMA =
VAR CurrentDate = MAX(Dim_Date[Date])
VAR PeriodDays = 50
VAR DateInPeriod =
    DATESINPERIOD(Dim_Date[Date], CurrentDate, -PeriodDays, DAY)
RETURN
CALCULATE(AVERAGE(Fact_Stock[Volume]), DateInPeriod)

```

PRICE TREND ANALYSIS:

The Close Price was used to see how the Shopify stock price moved over the years. The price chart shows the rise and fall of the stock across different periods, along with the moving average lines. The Daily Change % was also used to observe day-to-day movement in the price. Together, these show the overall direction and performance of the stock.





SLICERS AND DYNAMIC SELECTION:

Slicers were added to the dashboard so users can filter the data by date range or month. This lets users focus on the period they want to study, and the charts update automatically based on the selection. It makes the dashboard more interactive and user-friendly.

Month

- ☐ Apr
- ☐ Aug
- ☐ Dec
- ☐ Feb
- ☐ Jan
- ☐ Jul
- ☐ Jun
- ☐ Mar

CUSTOM TOOLTIP:

A custom tooltip page named Stock_Tooltip was created. When the user hovers over a chart, it shows the Open Price, High Price, Low Price, Close Price, Trading Volume and Daily Change % for that date. This provides detailed information without crowding the main dashboard.



CONCLUSION:

The Week 4 analysis made the Shopify stock dashboard more informative and interactive. The 20-Day and 50-Day moving averages helped in understanding trends in trading volume, and the comparison with daily volume showed the difference between short-term spikes and long-term patterns. The price analysis, slicers and custom tooltip made it easier to explore the data in detail. Overall, these features improved the dashboard and made the stock analysis easier to understand.